

NCT07562971: Focused Ultrasound Blood-Brain Barrier Opening in Pontine Diffuse Midline Glioma to Enhance Temozolomide Therapy: Pilot Feasibility Study (FIDES)

Gender: ALL

Age: 4 Years - MISSING

Criteria:

Inclusion Criteria:

- * Age $\geq$ 4 years.
- * Histologically/molecularly verified de novo pontine H3K27-altered diffuse midline glioma.
- * Main localization ('center of mass') in the brainstem. NB: some degree of extension beyond the brainstem, e.g. cerebellar peduncles, is allowed.
- * Karnofsky Performance Score (KPS) or Lansky Performance Score (LPS) of $\geq$ 70/ KPS or LPS 60 and WHO/ECOG performance status $\leq$ 2.
- * ASA-score of I-III.
- * Intention to treat with (TMZ chemo-) radiation and maintenance TMZ as per consensus of the local multidisciplinary tumor board.
- * Feasible to schedule the first Exablate BBBO procedure preferably within 4-6 weeks, acceptably within 12 weeks, after successful completion of radiotherapy/ concomitant TMZ-chemoradiation, defined as completed treatment as planned without reported CTCAEv6.0 grade 3-4 toxicities or, in case of reported CTCAEv6.0 grade 3-4 toxicities, the toxicities must be resolved to grade 2 prior to inclusion.
- * If on steroids, stable or decreasing dose for at least 7 days prior to inclusion.
- * Able to attend all study visits.
- * Able and willing to give informed consent or have a legal guardian who is able and willing to do so.

Exclusion Criteria:

- * Previous or ongoing participation in other clinical trials with other than standard-of-care tumor-directed treatment(s) for H3K27-altered DMG.
- * Multifocal or leptomeningeal metastasized disease. Multifocal disease is defined as multiple FLAIR-hyperintense lesions, separated by normal-appearing brain tissue, with or without gadolinium enhancement. Multiple enhancing regions within one continuous FLAIR lesion can be considered as unifocal disease.
- * Signs/symptoms of elevated intracranial pressure (ICP) (e.g. headache, vomiting, impaired vision/papilledema, impaired consciousness), with corresponding radiographic findings on MRI at time of screening.
- * Severe dysphagia with feeding tube dependency.
- * Evidence of acute clinically significant intracranial hemorrhage. NB: minimal hemorrhagic foci without obvious related clinical symptoms will not serve as grounds for exclusion.
- * Tumor not visible on any pre-therapy or post-radiation imaging.
- * Presence of extracranial / intracranial structures (e.g. metal prostheses, implants, calcifications) on pre-treatment CT-scan/ MRI-scan, significantly interfering with acoustic impedance as per judgement of the researchers.
- * Known co-occurring other malignancy that is progressing or has required active treatment within the past 3 years, with exception of: carcinomas in situ (CIS) and non-melanoma skin cancers.
- * Patients with right-to-left, bi-directional or transient right-to-left cardiac shunts.
- * Known LVEF $<$ 40 or unstable hemodynamics.
- * Severe hypertension, not adequately controlled with study compatible medication (Adults: RR systolic $>$ 180 and/ or RR diastolic $>$ 100; Children: $>$ p95 + 12mmHg).
- * History of bleeding disorder and/or coagulopathy.
- * Treatment with anti-coagulant therapy.
- * Severely impaired renal function; creatinine clearance $<$ 30 mL/ min.
- * Subjects with significant liver dysfunction; Child Pugh classification C.

- * Known diagnosis of active or untreated hepatitis B, hepatitis C, tuberculosis.
- * Any other illness or medical condition that in the investigator's opinion precludes participation in this study.
- * Pregnant or lactating women.
- * Expected uncontrollable therapy non-compliance/ non-cooperation that is likely to interfere with the study procedure, as per judgement of the investigators.
- * Head circumference ≤ 49 cm.
- * Weight ≥ 135 kg.
- * Patient ≥ 18 years old, who requires general anesthesia to undergo the Exablate BBBO procedure.
- * Contra-indication for MRI procedures.
- * Known sensitivity to gadolinium-based contrast agents.
- * Known sensitivity to the resonator agent (perflutren; Luminity).